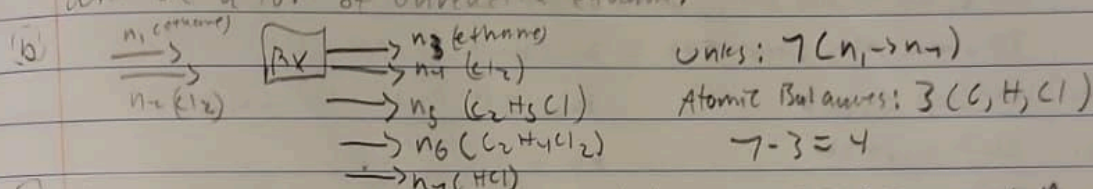


①

(a) In order to maximize selectivity, a low conversion of ethane should be targeted, since too much in the reactor would propel the second reaction. You will need a recycle stream and separator, since there will be a lot of unreacted ethane.



(c) Basis: 100 moles, moles ethane reacted = $100 \times .15 = 15 = n_{\text{mono}} + n_{\text{di}}$,
 $\frac{n_{\text{mono}}}{n_{\text{di}}} = 14$, $n_{\text{mono}} = 14$, $14n_{\text{mono}} + n_{\text{di}} = 15$, $n_{\text{di}} = 1 \text{ mol C}_2\text{H}_4\text{Cl}_2$,
 14 moles of $\text{C}_2\text{H}_5\text{Cl}$, total Cl_2 moles = $(1 \times n_{\text{mono}}) + (2 \times n_{\text{di}}) = 16 \text{ mol}$
 Feed ratio = $\frac{n_{\text{Cl}_2}}{n_{\text{C}_2\text{H}_6}} = \frac{16}{100} = 0.16$, Fractional Yield = $\frac{14}{16} = 87.5\%$

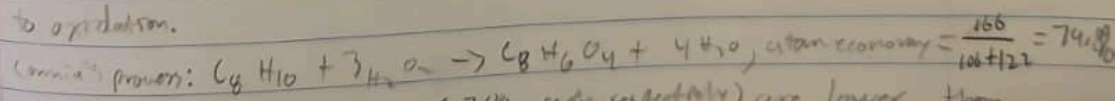
(d) It likely means there was further chlorination, getting $\text{C}_2\text{H}_3\text{Cl}_3$ in the product.

(2) Assume 1 mol of fed: $x = \text{C reacted}$, $n_6 = 1 - x$, $n_7 = 5 - 3x$, $n_8 = 3x$, $n_{\text{H}_2} = 7x$
 total moles = $6 + 6x$, $\frac{7x}{6 + 6x} = 0.54$, $x = 0.86$, $X_g = 86.7\%$
 $Q_p = \frac{(0.31 \times 1.2)(0.54 \times 1.2)^{6+6x}}{(0.09 \times 1.2)(0.16 \times 1.2)} = 1.65$, the reactor is not at equilibrium.

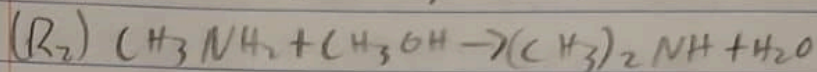
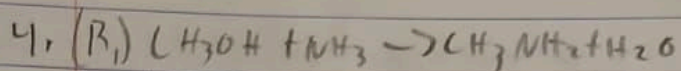
Either increase temperature (since it should be endothermic) or increase residence time.

$$\frac{(3x)^3(7x)^7}{(1-x)(5-3x)} \times \frac{P^6}{(6+6x)^6} = 55, \text{ Solver says } 0.9 \text{ equilibrium (conversion)}$$

3. p-xylene + $\text{O}_2 \rightarrow \text{TA} + \text{H}_2\text{O}$, high selectivity, but loses 10% acetic acid to oxidation.



Convent's yields and selectivity (70%, 90% respectively) are lower than the conventional process. Also, H_2O_2 is more expensive.



Total feed = 200 mol

$200 \times 0.2 = 40 \text{ mol NH}_3$, $200 \times 0.094 = 18.8 \text{ mol MOH}$, $200 \times 0.186 = 37.2 \text{ mol MA}$, $200 \times 0.04 = 8 \text{ mol DMA}$, $\text{H}_2\text{O} = 82.2 \text{ mol}$

a. $\frac{100 - 18.8}{100} = 0.812$

$K_{y1} = \frac{(0.186)(0.411)}{(0.044)(0.2)} = 4.07$

b. $\frac{37.2}{100} = 0.372$

$K_{y2} = \frac{(0.104)(0.411)}{(0.186)(0.064)} = 2.5$

c. $37.2 / 21.8 = 1.71$

d. $100 - 40 = 60 \text{ g mol/hr}$

e. 21.8 g mol/hr

The reactions are kinetically at equilibrium